

MAPURANA



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MARKING SCHEME

NOVEMBER 2014

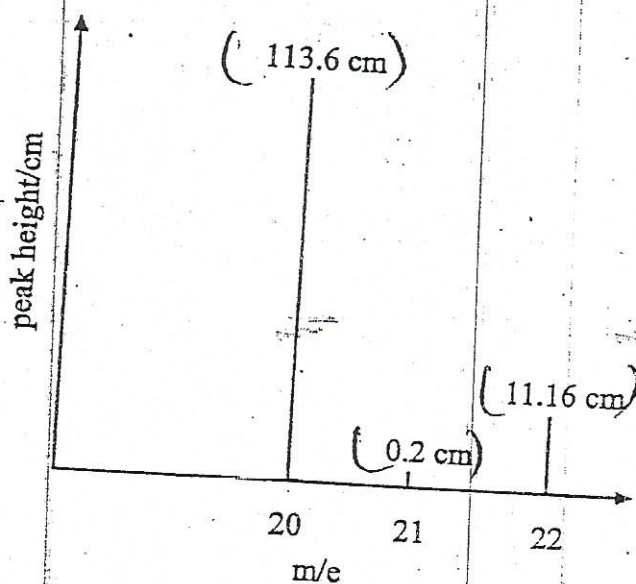
CHEMISTRY 9189/1

(a) Relative isotopic mass = $\frac{\text{mass of one atom of an isotope of an element}}{\frac{1}{12} \text{ mass of } ^{12}\text{C} \text{ atom}}$ /AW1 [1]

(b) (i) For ^{20}Q : $0.16\% \rightarrow 0.2 \text{ cm}$
 $90.91\% \rightarrow \left(\frac{90.91}{0.16} \times 0.2 \right) = 113.64 \text{ (cm)} [1]$

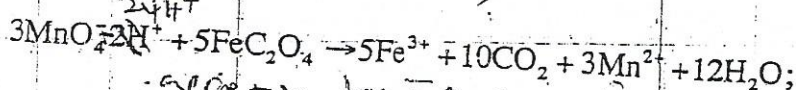
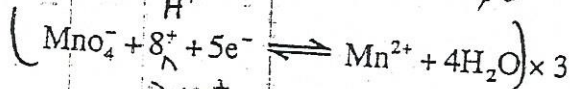
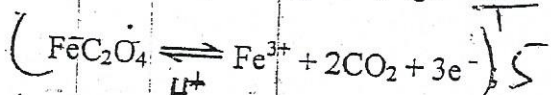
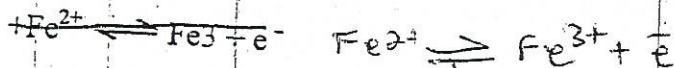
For ^{22}Q : $\frac{8.93}{0.16} \times 0.2 = 11.16 \text{ (cm)} [1]$

(ii)



Sketch 1 mark.

(iii)
$$\text{Ar}(\text{Q}) = \frac{(90.91 \times 20) + (0.16 \times 21) + (8.93 \times 22)}{100} = \frac{20 \times 113.64 + 21 \times 0.2 + 22 \times 11.16}{100} = 20.18; \quad 20.20$$
 [1]



also deduced equation [1]

(ii)

$n(\text{FeC}_2\text{O}_4) \text{ in } 100 \text{ cm}^3 \text{ of solution} = \frac{15}{\text{Mr}(\text{FeC}_2\text{O}_4)} = \frac{15}{143.8} = 0.1043 \text{ moles}$ [1]

Allow 1 mark on this question.

$$n(\text{FeC}_2\text{O}_4) \text{ in } 25 \text{ cm}^3 = \frac{1}{4} \times 0.01043$$

$$= 0.00261 \text{ moles;}$$

Use of mole ratio $\text{MnO}_4^- : \text{FeC}_2\text{O}_4 = 3 : 5$;

allow rcf.

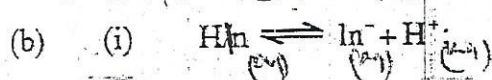
$$n(\text{MnO}_4^-) = \frac{0.00261 \times 3}{5} = 0.00157 \text{ moles;}$$

$$[\text{MnO}_4^-] = \frac{0.00157 \times 1000}{15} = 0.105 \text{ mol dm}^{-3}$$

[Total: 12]

(a) A weak acid whose conjugate base has a different colour from it; AW

Substance that changes colour with change in pH



(R) wrong S.S.
(A) wrong S.S.

(ii) $K_{\text{In}} = \frac{[\text{H}^+][\text{In}^-]}{[\text{HIn}]}$

(c) (i) $\text{pH} = \text{pK}_{\text{In}} = -\lg K_{\text{In}} = -\lg(1.2 \times 10^{-4}) = 3.9$; (Ans to 3.9)

(ii) Strong acid weak base titrations; / Accept names of strong acid and weak bases

(iii) $\text{pH} = -\lg \sqrt{K_{\text{In}}[\text{HIn}]}$ $[\text{H}^+] = \sqrt{1.2 \times 10^{-4} \times 0.2}$
 $= -\lg \sqrt{1.2 \times 10^{-4} \times 0.2} = 0.0049$
 $= 2.3$; $\text{pH} = -\lg(0.0049) = 2.3$

Calculation;

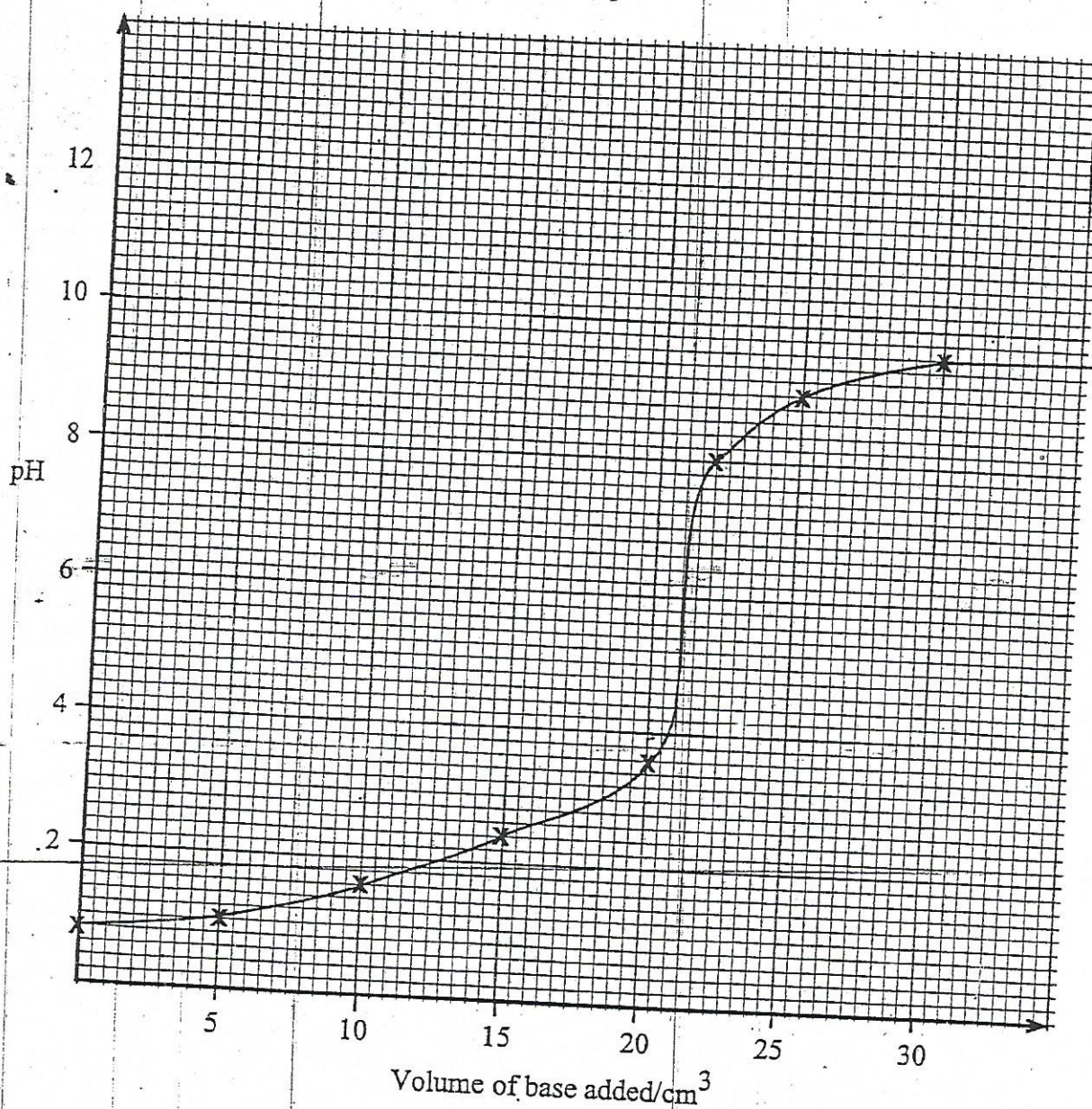
Answer;

(iv) The two to three drops of Indicator added during titrations is too minute to affect the pH of the solution; AW;

(d) (i) graph

scale - [1]
 plots - [1]
 curve - [1]

*Score changed by
 Inspected.*



(ii)

End point pH = ~~6.1~~
 Shown on graph

~~5.8 - 6.3~~

[1]

[Total: 12]

Route and path.

(a) (i) Steps showing actual changes during a chemical reaction/AW;

[1]

(ii) To be able to control speeds of chemical reactions; To understand reaction mechanisms; Ensure safety of workers/save time/save resources / *end.*

[1]

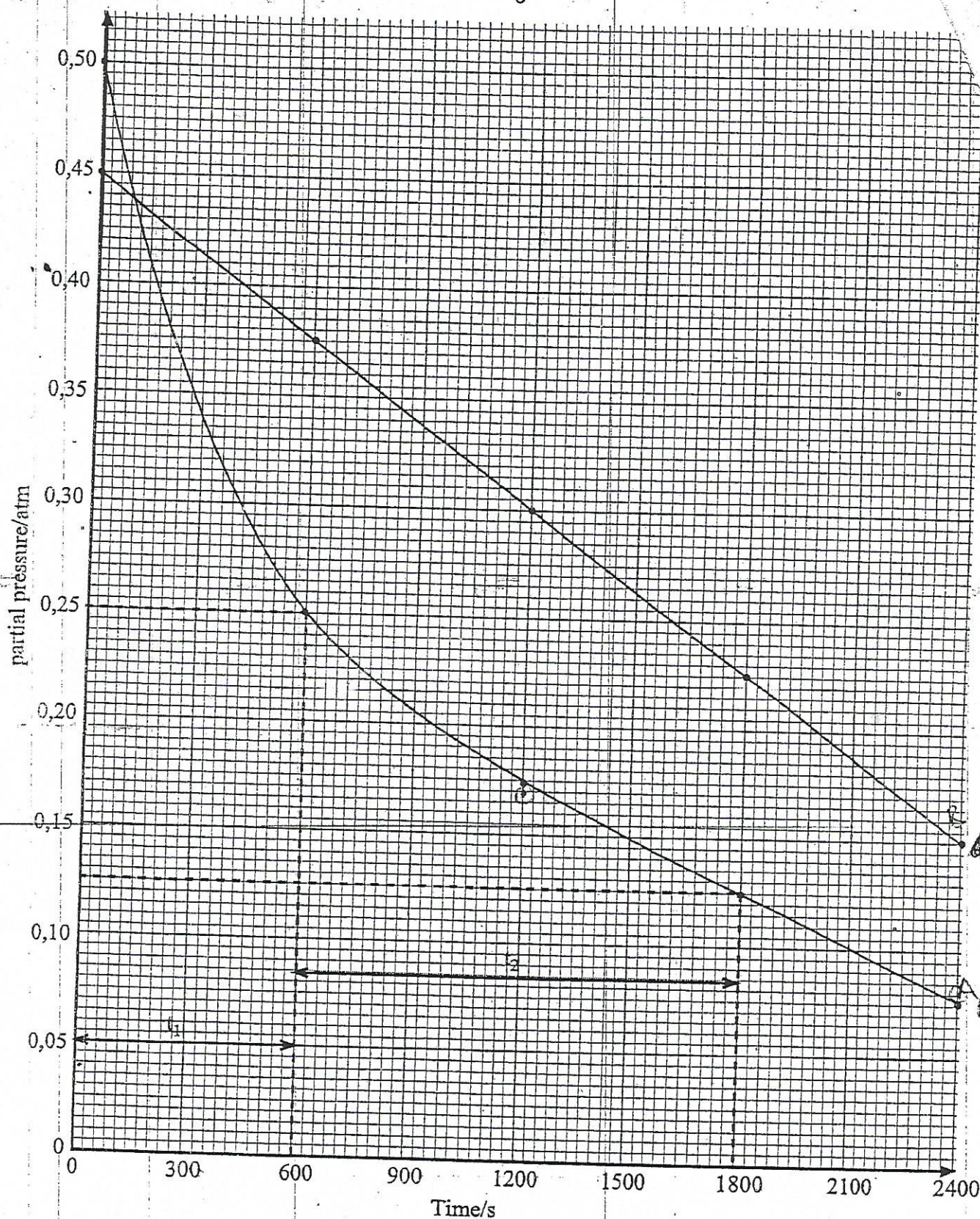
(b) (i) Scale *+ labels* [1]

Plots [1] *+ 0.5*

Line [1]

Curve [1]

lower at least half the space



- (ii) I Order with respect to A is 2
half-life not constant; *Shown on graph.*
- II Order with respect to B is 0

[1]

[1]

~~For~~

- For students who do not use a graph don't award for marks for plots and scale.

- Graphite without hand-drawn

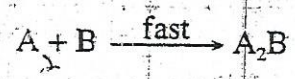
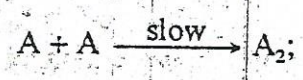
Answer to Straight line graph;

[1]

(ii) ~~Rate = k[A]~~

~~[1]~~

(iii) Mechanism



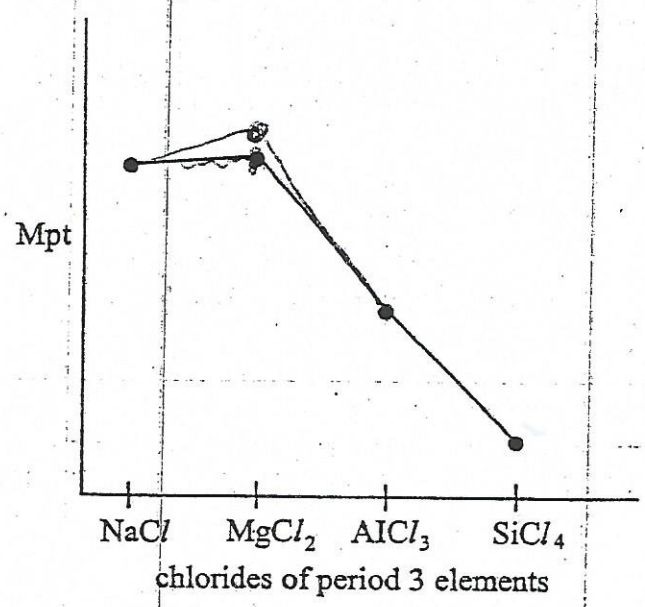
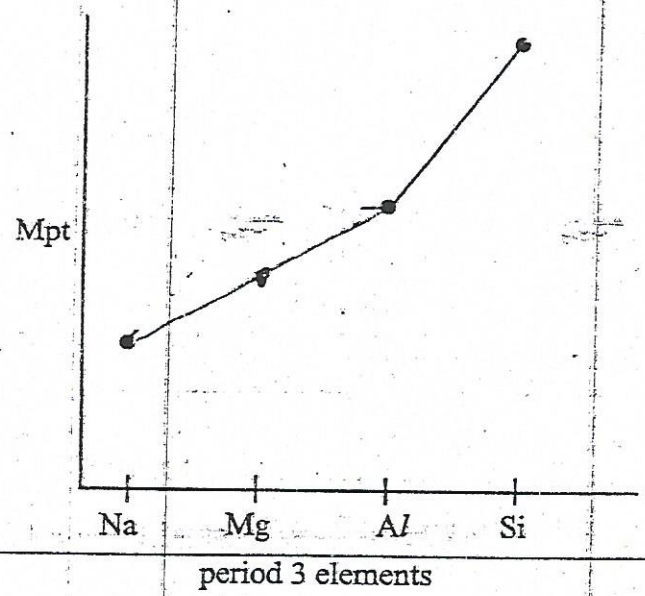
Salvage (1)
for both equations
but not
inserted.

[1]

[1]

[Total: 12]

4 (a) (i)



m₄Cl₂ 15/12/2011
2/1/11

1-13

- (ii) - Na - Al are (giant metallic) strength of metallic bond increases with increase in number of delocalised electrons; [1]
- Si is giant molecular more energy needed to break the strong covalent bonds; [1]
- Mg^{2+} has higher charge density than Na^+ [1]
- NaCl and MgCl_2 are (giant ionic) Ions are held by strong electrostatic forces of attraction, /AN [1]
- AlCl_3 and SiCl_4 are simple molecular, covalent molecules are held by weak van der Waal's forces; [1]
- AlCl_3 exist as a dimer so has larger size and stronger VDW than SiCl_4 ; /AN [1]

- (b) (i) X is sulphur/S; S for S [1]
- (ii) Peak due to molecule ion S_8^+ ; Sulphur charged. [1]

- (iii) I. $\text{X}_2\text{O}_3^{2-} + 4\text{Cl}_2 + 5\text{H}_2\text{O} \rightarrow 8\text{Cl}^- + 2\text{XO}_4^{2-} + 10\text{H}^+$; [1]
- II. $2\text{X}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{X}_4\text{O}_6^{2-} + 2\text{I}^-$ (A) S for X. [1]
3. $\text{X}_2\text{O}_3 + 2\text{H}^+ \rightarrow \text{XO}_2 + \text{X} + \text{H}_2\text{O}$; [1]

[Total: 12]

- 5 (a) (i) 1. A lot of energy is needed to break the $\text{N}=\text{N}$ bond of N_2 [1]

2. P_4 is a larger molecule than N_2 [atomity of P is 4 while that of N is 2,] [1]
- There are more electrons in P_4 units than on N_2 molecules hence Stronger van der Waal's Forces; [1]

- (ii) 1. $4\text{P} + 3\text{O}_2 \rightarrow 2\text{P}_2\text{O}_3/\text{P}_4\text{O}_6$; (A) P_4 [1]
- $4\text{P} + 5\text{O}_2 \rightarrow 2\text{P}_2\text{O}_5/\text{P}_4\text{O}_{10}$
2. $\text{P}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{PO}_3$; OR $\text{P}_4\text{O}_6 + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_3$
- $\text{P}_2\text{O}_5 + 3\text{H}_2\text{O} \rightarrow 2\text{H}_3\text{PO}_4$; $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$ [1]

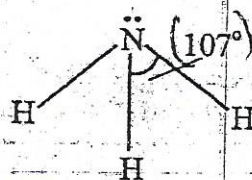
- (b) (i) As a refrigerant; making detergents; Manufacture of nitric acid; any two amines, amides, nylon, paints, dyes, explosives [1]
- (ii) $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$; [1]

ignore S.S

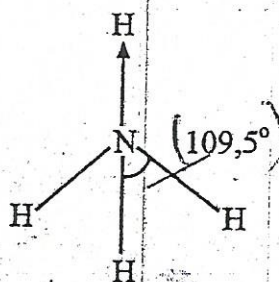
(R) wrong S.S

Salvage (A) for the equations but wrong S.S.

(c) (i)



(Trigonal) pyramidal ammonia



tetrahydronium ion [2]

(ii)



1. has a characteristic smell, liquid does, not conduct electricity

No smell, conducts in molten form, aqueous soln.

Low Mp point
Bpt

high Mp point

2. basic / A

acidic / A [2]

[Total: 12]

(E) should show comparison
(E) negative answering

(a) (i)

B

electron withdrawing NO₂ group weakens O-H bond / A [1]
stabilises the anion

(ii)

C

has a lone pair of electrons on N which it can use to accept a proton/H⁺ ion; [1]

(iii)

D

lone pair on N delocalised over amide bond; [1]

E

no lone pair on N all e⁻ used in bonding; [1]

C

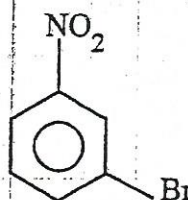
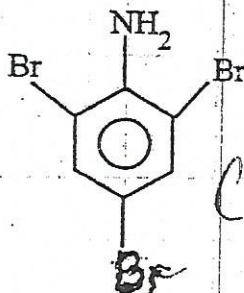
no reason no mark

E

1.

Br_{2(aq)}, r.t.p/ordinary conditions
(without light) (1)Br_{2(aq)}, A/Br₃/
FeBr₃/Fe catalyst
in the dark (1) [2]

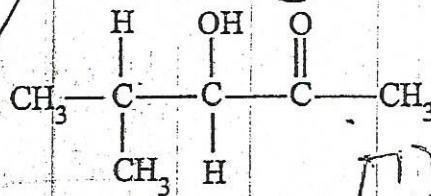
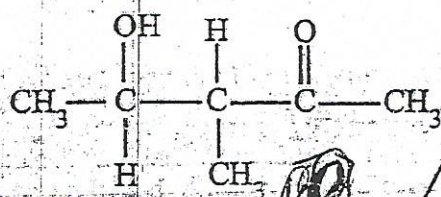
specific catalyst



2.

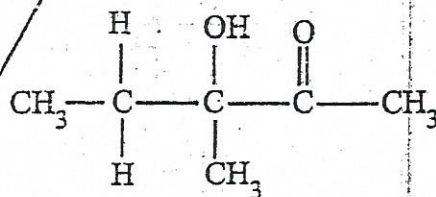
max 4

(b) (i) Geometric/cis/trans; optical; [2]



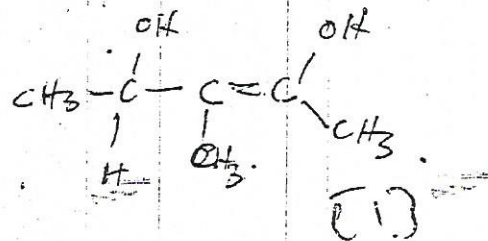
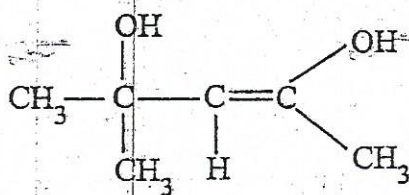
(ii)

1.

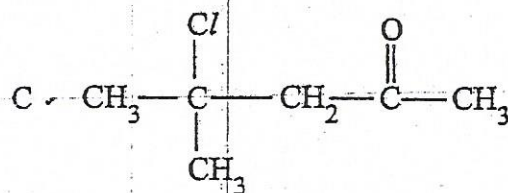
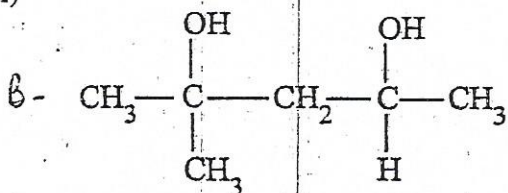


optical

2.

(c) (i) w - conc H_2SO_4 at 170°C /conc H_3PO_4 at 300°C /
Hot Al_2O_3 /ceramic; [1]Y - $\text{NaOH}_{(\text{aq})} + \text{I}_{2(\text{aq})}$ heat/warm; [1]

(ii)

(iii) w line ~~part~~ solid ~~part~~ / two separate layers. (13)

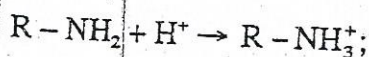
- (iv) Insoluble halogeno compound formed from soluble alcohol;

Formation of $\text{R}(\text{ON})_2$

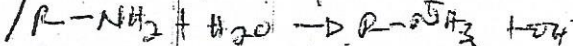
[1]
[Total: 12]

- (a) (i) Solubility is due to formation of hydrogen bonds (between the amine group and water molecules) / *low* [1]
 (ii) Higher members have larger hydrophobic parts hence less soluble; *in polar protic* [1]
 (iii) Have stronger van der Waal's forces hence less volatile; [1]

- (b) (i) Amines have a lone pair of electrons on the N atom which they use to accept a proton; *⊗ ≡* [1]



⊗ square for Amine

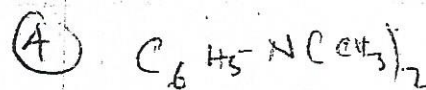
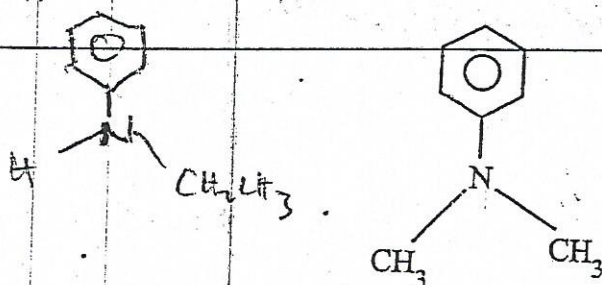


- (ii) The higher the pK_b , the lower the K_b , and the less basic the amine $\Rightarrow$ phenylamine less basic than methylamine; [1]

The lone pair of electrons on N in phenylamine is delocalised into the Ring hence not always available to accept a proton; [1]

The methyl group in methyl amine has a positive inductive effect hence increases the electron density on N; [1]

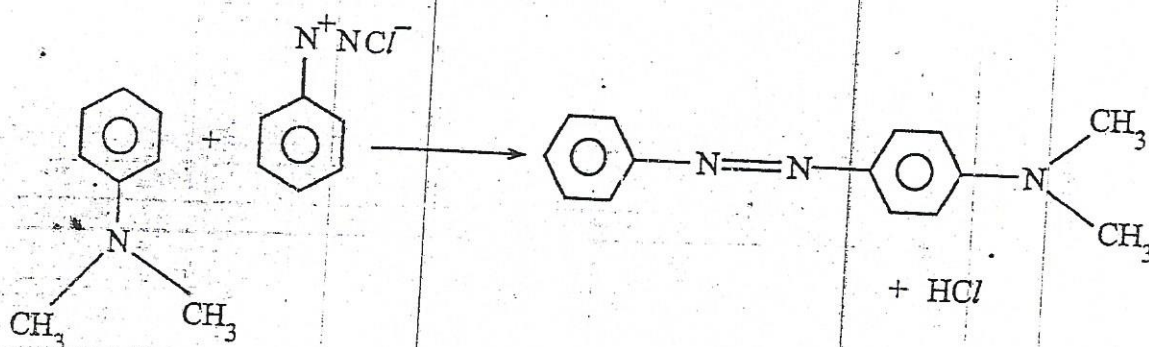
- (c) (i)



- (ii) $T < 10^\circ\text{C}$ (mild alkali; NaOH or KOH) [1]

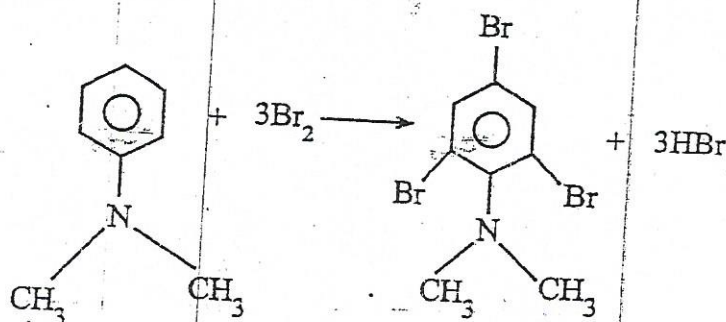
13
Allied eff for wrong structure

1.



[1]

2.



[1]

[Total: 12]